

Computer and Cyber Crimes

What is Cybercrime?

- Using the Internet to commit a crime.
 - Identity Theft
 - Hacking
 - Viruses
- Facilitation of traditional criminal activity
 - Stalking
 - Stealing information
 - Child Pornography

Cybercrime Components

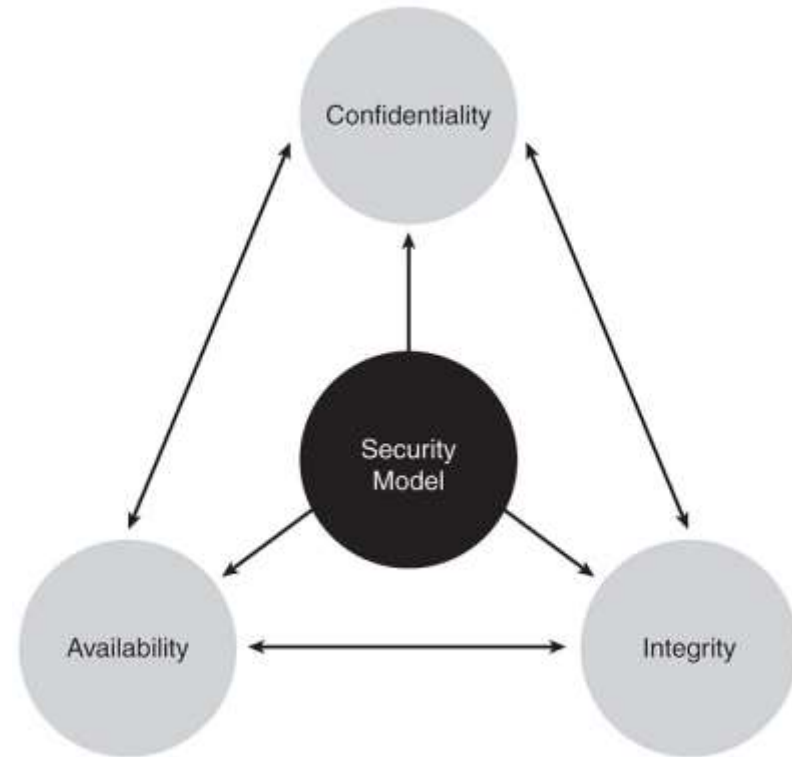
- Computers
- Cell Phones
- PDA's
- Game Consoles

High-Profile Cybercrime related Cases

- TJ Maxx data breach
 - 45 million credit and debit card numbers stolen
- Kwame Kilpatrick
 - Cell phone text messages
- BTK Serial Killer
- Kevin Mitnick

Computer Security

- Confidentiality
 - Only those authorized to view information
- Integrity
 - Information is correct and hasn't been altered by unauthorized users or software
- Availability
 - Data is accessible to authorized users



Categories of cyber crime

We can categorize cyber crime in two ways.

- **The computer as a target** :- using a computer to attacks other computer, e.g. Hacking, virus/worms attacks, Dos attack etc.
- **The computer as a weapon** :- using a computer to commit real world crime e.g. cyber terrorism, credit card fraud and pornography etc.

Top Cyber Crimes that Attack Business

Spam

Viruses/Worms

Industrial Espionage and Hackers

Wi-Fi High Jacking

Spam is Hostile

- You pay for Spam, not Spammers
 - Email costs are paid by email recipients
- Spam can be dangerous
 - Never click on the opt-out link!
 - May take you to hostile web site where mouse-over downloads an .exe
 - Tells spammers they found a working address
 - They won't take you off the list anyway
- What should you do?
 - Filter it out whenever possible
 - Keep filters up to date
 - If you get it, just delete the email



Viruses and Worms

- Different types of “ailments”
- Viruses
 - software that piggybacks on other software and runs when you run something else
 - Macro in excel, word
 - Transmitted through sharing programs on bulletin boards
 - Passing around floppy disks
 - An .exe, .com file in your email
- Worms
 - software that uses computer networks to find security holes to get in to your computer – usually in Microsoft OS!! But worm for MAC was recently written



Types of cyber crime

- **CYBER TERRORISM** : Terrorist attacks on the Internet is by distributed denial of service attacks, hate websites and hate E-mails , attacks on service network etc.
- **SOFTWARE PIRACY** : Theft of software through the illegal copying of genuine programs or the counterfeiting and distribution of products intended to pass for the original.
- **CHILD PORNOGRAPHY** : The Internet is being highly used by its abusers to reach and abuse children sexually, worldwide. As more homes have access to internet, more children would be using the internet and more are the chances of falling victim to the aggression of Pedophiles.

Pedophiles

- Internet allows them to
 - Instant access to other predators worldwide;
 - Open discussion with children
 - Support from other pedophiles
 - Disguise their identities
- Pedophile organizations include
 - NAMBLA (The North American Man-Boy Love Association) and
 - PAN (Pedophile Alert Network) in the Netherlands.
 - Members receive monthly magazines and newsletters that include seduction techniques and advice on avoiding detection and prosecution. Pedophiles meet each other via the Internet where they swap methods, success stories, even names, descriptions, and images of children.

Hackers are Everywhere

- Stealing data
 - Industrial Espionage
 - Identity theft
 - Defamation
- Deleting data for fun
 - A lot of bored 16 year olds late at night
- Turning computers into zombies
 - To commit crimes
 - Take down networks
 - Distribute porn
 - Harass someone
- Ethical/white hat hackers exist too
 - Help break into networks to prevent crimes

Denial of Service Attack

- An act by the criminal, who floods the bandwidth of the victim's network or fills their e-mail box with spam mail depriving him of the services he is entitled to access or provide

Virus Dissemination

- Malicious software that attaches itself to other software.
- (Virus, Worms, Trojan Horse, Time Bomb, Logic Bomb)

Software Piracy

- Theft of Software through the illegal copying of genuine programs or the counterfeiting and distribution of products intended to pass for the original
 - Examples(Pirate Bay, Bootlegs, etc)
 - End user copying, Downloads

IRC Crime

Internet Relay Chat (IRC) servers have chat rooms in which people from anywhere the world can come together and chat with each other

- Criminals use it for meeting coconspirators.
- Hackers use it for discussing their exploits / sharing the techniques
- Pedophiles use chat rooms to allure small children
- Cyber Stalking - In order to harass a woman her telephone number is given to others as if she wants to befriend males

Credit Card Fraud

- If electronic transactions are not secured the credit card numbers can be stolen by the hackers who can misuse this card by impersonating the credit card owner

NET Extortion

- Copying the company's confidential data in order to extort the company for a huge amount

Phishing

- A technique of pulling out confidential information from the bank/financial accounts by deceptive means

EX: of Phishing Email

From: *****Bank [mailto:support@*****Bank.com]

Sent: 08 June 2004 03:25

To: India

Subject: Official information from ***** Bank

Dear valued ***** Bank Customer!

For security purposes your account has been randomly chosen for verification. To verify your account information we are asking you to provide us with all the data we are requesting. Otherwise we will not be able to verify your identity and access to your account will be denied. Please click on the link below to get to the bank secure page and verify your account details. Thank you.

https://infinity.*****bank.co.in/Verify.jsp

***** Bank Limited

Spoofing

- Getting one Computer on the network to pretend to have the identity of another computer, usually one with special access privileges, so as to obtain access to the other computers on the network.

Cyber Stalking

- The criminal follows the victim by sending emails, entering the chat rooms as the person being stalked frequently

Cyber Defamation

- The Criminal sends emails containing defamatory matters to all concerned of the victim or post the defamatory matters on a website..
- (disgruntled employee may do this against boss,, ex-boys friend against girl,, divorced husband against wife etc)

Threatening

- The criminal sends threatening email or comes in contact in the chat rooms with victim
- (Anyone disgruntled may do this against boss, friend or official)

Salami Attack

- In such crime criminal makes insignificant changes in such a manner that such changes would go unnoticed.
- Criminal makes such program that deducts small amount like Rs. 2.50 per month from the account of all the customer of the Bank and deposit the same in his account. In this case no account holder will approach the bank for such small amount but criminal gains huge amount.

Sale of Narcotics

- Sale and Purchase through the net
- Websites offer sales and shipment of contraband drugs
- May use hidden messages to sell the drugs

Protect your Computers!

- Use anti-virus software and firewalls - keep them up to date
- Keep your operating system up to date with critical security updates and patches
- Don't open emails or attachments from unknown sources
- Use hard-to-guess passwords. Don't use words found in a dictionary. Remember that password cracking tools exist
- Back-up your computer data on disks or CDs often
- Don't share access to your computers with strangers
- If you have a wi-fi network, password protect it
- Disconnect from the Internet when not in use
- Reevaluate your security on a regular basis
- Make sure your employees and family members know this info too!

Computer Forensics

- What is it?
 - an autopsy of a computer or network to uncover digital evidence of a crime
 - Evidence must be preserved and hold up in a court of law
- Growing field – Many becoming computer forensic savvy
 - FBI, State and Local Police, IRS, Homeland Security
 - Defense attorneys, judges and prosecutors
 - Independent security agencies
 - White hat or Ethical Hackers
 - Programs offered at major universities such as URI

Uncovering Digital Evidence

Smart Criminals don't use their own computers

- Floppy disks
- Zip/Jazz disks
- Tapes
- Digital cameras
- Memory sticks
- Printers
- CDs
- PDAs
- Game boxes
- Networks
- Hard drives



Digital Evidence

***Not obvious.....it's most likely hidden on purpose
or needs to be unearthed by forensics experts***

Criminals Hide Evidence

- Delete their files and emails
- Hide their files by encryption, password protection, or embedding them in unrelated files (dll, os etc)
- Use Wi-Fi networks and cyber cafes to cover their tracks

Forensics Uncover Evidence

- Restore deleted files and emails – they are still really there!
- Find the hidden files through complex password, encryption programs, and searching techniques
- Track them down through the digital trail - IP addresses to ISPs to the offender

The Crime Scene

(with Computer Forensics)

- Similar to traditional crime scenes
 - Must acquire the evidence while preserving the integrity of the evidence
 - No damage during collection, transportation, or storage
 - Document everything
 - Collect everything the first time
 - Establish a chain of custody
- But also different.....
 - Can perform analysis of evidence on exact copy!
 - Make many copies and investigate them without touching original
 - Can use time stamping/hash code techniques to prove evidence hasn't been compromised



CYBER SECURITY

Cyber Security : Cyber Security involves protection of sensitive personal and business information through prevention, detection and response to different online attacks. Cyber security actually preventing the attacks, cyber security.

Privacy Policy : Before submitting your name , e-mail , address, on a website look for the sites privacy policy.

Keep Software Up to Date: If the seller reduces patches for the software operating system your device ,install them as soon as possible .Installing them will prevent attackers form being able to take advantage. Use good password which will be difficult for thieves to guess. Do not choose option that allows your computer to remember your passwords.

- **DISABLE REMOTE CONNECTIVITY** : Some PDA's and phones are equipped with wireless technologies, such as Bluetooth, that can be used to connect to other devices or computers. You should disable these features when they are not in use.

ADVANTAGES OF CYBER SECURITY

- The cyber security will defend us from critical attacks.
- It helps us to browse the site, website.
- Internet Security process all the incoming and outgoing data on your computer.
- It will defend us from hacks and virus.
- Application of cyber security used in our PC needs update every week

SAFETY TIPS TO CYBER CRIME

- Use antivirus Software
- Insert Firewalls
- Uninstall unnecessary software
- Maintain backup
- Check security settings